

Physician-Authored Evaluation of Patient-Facing Artificial Intelligence in Emergency Triage

A Scientific Introduction to the Research Protocol

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Version 3 — August 2026, revised 21 August 2026. This revision incorporates the 2026 evidence base (Ramaswamy et al., Nature Medicine; the evaluation-format dispute; the NOHARM benchmark and randomized study) and the methods hardening of Amendment 1 and Addendum A. The 21 August revision adds the deployment safety classification of Sheng et al. (NEJM AI, 2026) — a proposed framework — together with the study limitations it makes explicit, and places the work within the current regulatory architecture. Preprint citations are identified as such throughout.

The Problem: A Safety Vacuum at the Point of First Contact

Every day, millions of people experiencing symptoms they do not fully understand type those symptoms into an artificial intelligence system and ask a question that carries life-or-death weight: *Do I need to go to the emergency room?* The large language models answering that question — ChatGPT, Claude, Gemini, and their successors — were not designed as triage instruments, were never validated against emergency medicine standards, and operate without any regulatory framework governing their safety in this role. Yet they function as de facto triage systems for a growing share of the population, issuing recommendations that directly influence whether a patient with an evolving ST-elevation myocardial infarction drives to the emergency department or takes an antacid and goes to bed.

A framework published this month in *NEJM AI* makes the mismatch explicit. Sheng et al. propose four deployment-specific safety levels for medical AI and place “autonomous emergency department triage/discharge” and “autonomous denial or delay of specialist referral” at S4 — the red line, where the required clinician action is “do not permit autonomous action” and the deployment is to be treated as boxed warning-level risk. Clinician-reviewed triage and care-navigation support sits two tiers lower, at S2, precisely because a clinician stands between the model’s output and the patient. Consumer patient-facing triage AI has no clinician in that loop. On the framework’s own terms, the configuration now running at population

scale is an S4 configuration operating without S4 safeguards — and it has never been classified at anything, because no one has classified it at all. The framework is a proposal and carries no data of its own; what it supplies is the vocabulary for why the measurement is owed. This protocol is the instrument for making that classification empirically, before rather than after the harm.

The clinical AI safety literature has not kept pace with this reality. The dominant evaluation paradigm tests whether language models can answer medical knowledge questions correctly — board-style multiple-choice items from USMLE, MedQA, MedMCQA, and similar benchmarks. These assessments measure the wrong construct. A model that achieves 96% accuracy on USMLE Step 1 questions, as OpenAI's o1 has done, demonstrates recall of medical facts under examination conditions. It does not demonstrate the capacity to recognize that a 74-year-old diabetic man describing fatigue, nausea, and exertional dyspnea — without chest pain — requires immediate emergency evaluation for an atypical acute coronary syndrome. The gap between knowledge retrieval and triage safety is not a nuance; it is the entire clinical question. That gap has now been measured directly: in the NOHARM benchmark, models' clinical-safety performance correlated only moderately ($r = 0.61$ – 0.64) with their scores on existing AI and medical knowledge benchmarks — a model can excel on accuracy evaluations and still produce severely harmful recommendations at nontrivial rates (Wu et al., preprint).

Emergency triage operates under a principle of asymmetric harm that has no parallel in diagnostic accuracy benchmarks. When a triage decision errs toward caution — sending a patient with benign chest wall pain to the emergency department — the consequence is inconvenience, unnecessary resource utilization, and modest cost. When a triage decision errs in the other direction — reassuring a patient whose thunderclap headache is a subarachnoid hemorrhage that they likely have a migraine — the consequence may be death within hours. This asymmetry is not an incidental feature of triage; it is the organizing principle of the Emergency Severity Index, the standard triage instrument used across American emergency departments. The American College of Surgeons codifies this asymmetry quantitatively: trauma systems target an undertriage rate below 5% while tolerating overtriage rates as high as 35%. Any evaluation framework that treats these two error types symmetrically — as nearly all existing AI medical benchmarks do — fundamentally misrepresents the clinical reality of triage decision-making.

This study was designed to close that gap. It asks a single, operationally precise question: when patients describe symptoms in everyday language to a large language model, does the model reliably recognize presentations requiring emergency care? The answer to this question has immediate

implications for patient safety, clinical policy, and the regulatory trajectory of consumer-facing health AI.

2026: The Failure Is No Longer Hypothetical

Three converging lines of evidence now define the safety problem this trial measures.

First, the failure phenotype has surfaced in the wild. *Winters v. OpenAI* (San Francisco Superior Court; complaint filed July 22, 2026) alleges that a consumer chatbot met an evolving venous thromboembolism with reassurance, characterized recurrent dizziness and blood-pressure instability as minor, invented a clinical threshold for seeking care, and advised immobility — with a fatal massive pulmonary embolism attributed in part to the advice given. These are allegations in active litigation, not findings; but the alleged mechanism is a disposition failure, not a diagnostic one, and it is precisely the failure class this protocol is built to measure.

Second, the phenotype has been registered under controlled conditions in the deployed product. In a landmark independent evaluation, Ramaswamy and colleagues at Mount Sinai stress-tested ChatGPT Health — OpenAI’s consumer health product, on its live interface — with 60 clinician-authored vignettes across 21 clinical domains under 16 factorial conditions (Ramaswamy et al., *Nature Medicine* 2026). Under that constrained testing, the system undertriaged 51.6% of gold-standard emergencies, shifted toward less urgent care when family members minimized symptoms (odds ratio 11.7, 95% CI 3.7–36.6), and activated crisis-intervention safeguards unpredictably across suicidal-ideation presentations. Within weeks, the measurement itself became contested: a partial replication argued that the study’s forced-choice elicitation format, rather than model capability, manufactured much of the headline failure rate (Fraile Navarro, Magrabi & Coiera, preprint). The dispute is examined in detail below; its existence is itself part of the problem this protocol addresses — the field disagrees not only about how safe these systems are, but about how safety should be measured at all.

Third, the deployment interface degrades whatever capability the model has in isolation. In Bean et al.’s randomized evaluation (*Nature Medicine* 2026), models that identified conditions at 94.9% when tested alone fell below 44.2% disposition accuracy in lay users’ hands — no better than control. Physician-side gains from LLM assistance have been similarly marginal (Goh et al., *JAMA Network Open* 2024; Healy et al., *Diagnosis* 2026), and in 2026 the NOHARM randomized study measured the mechanism at scale (Wu, Nateghi Haredasht et al., preprint): 101 attending physicians given AI assistance improved modestly over conventional resources (47.3% vs 42.2% severity-weighted performance) yet remained far below frontier models working alone, in large part because they omitted

appropriate recommendations their own AI assistant had surfaced. The interface — human or lay — gives back a substantial share of what the model can do.

An allegation, a contested measurement, and an interaction gap — and still no standard instrument. No published evaluation combines physician-authored, contamination-audited scenarios; free-response elicitation with deterministic escalation detection; an independent physician answering benchmark; severity-weighted disposition scoring; and a human-AI interaction arm. This protocol supplies that instrument.

Scientific Foundations: Why Existing Benchmarks Cannot Answer This Question

The inadequacy of current evaluation methods for this purpose is not merely a matter of scope or difficulty level. It reflects four structural incompatibilities between existing benchmarks and the construct of triage safety, each supported by a distinct body of evidence.

First, the format is wrong. The overwhelming majority of medical AI benchmarks use multiple-choice questions. Research published in 2025 in *Scientific Reports* (the FreeMedQA benchmark, Singh et al.) demonstrated that large language models exploit answer options independent of clinical knowledge, achieving performance on identical medical content that drops by 37 to 46 percentage points when converted from multiple-choice to free-response format. GPT-4o's performance fell 37.5 points; Llama-3-70B dropped 46.6 points. With question stems completely masked, GPT-4o still achieved 37.3% on multiple-choice items against a 25% random baseline — direct evidence of systematic answer-option exploitation. The CRAFT-MD study in *Nature Medicine* (Johri et al., 2025) confirmed this pattern across all tested models. When a patient types symptoms into a chatbot, no answer options are presented. The model must generate its recommendation from an unconstrained output space. Any evaluation that presents answer choices is testing a fundamentally different cognitive task than the one the model performs in deployment. The 2026 dispute over the ChatGPT Health evaluation, discussed below, is this incompatibility playing out in public — with constrained-format effects large enough to move a headline safety estimate.

Second, the clinical construct is wrong. Benchmarks such as MedXpertQA, AMIS, and MedQA test specialist-level diagnostic reasoning with access to complete clinical workups — serial laboratory values, imaging results, hospital course data, treatment responses observed over days. Triage decisions are made in two to five minutes with presenting information only: chief complaint, brief history, vital signs, and clinical appearance. A model that correctly identifies fulminant liver failure from five days of serial hepatic function panels demonstrates specialist diagnostic

acuity. It demonstrates nothing about whether that same model can correctly assign an ESI Level 1 to an undifferentiated patient presenting with altered mental status and hypotension before any laboratory data exist. The informational constraints are categorically different, and performance under one set of constraints does not predict performance under the other.

Third, the contamination threat is unaddressed. Until 2026, every published LLM triage study used either the ESI Implementation Handbook's freely available case library or synthetic vignettes derived from published sources. The ESI v4 handbook has been available as a downloadable PDF from AHRQ since 2012 and is reproduced across Quizlet, Studocu, Course Hero, and educational platforms worldwide. A 2025 study explicitly described continued pretraining of Llama-3.1 on the ESI handbook content for ten epochs. The recent ChatGPT Health evaluation (Ramaswamy et al.) is the notable exception — its 60 vignettes were clinician-authored de novo, an approach this protocol shares and extends. No published evaluation, however, has embedded contamination-*detection* items to measure memorization effects directly rather than assume their absence — a striking methodological gap in a field where the ICML 2025 study by Sun et al., testing 20 decontamination strategies across 10 models and 5 benchmarks, concluded that no existing surface-level perturbation strategy reliably balances fidelity and contamination resistance. If frontier models have memorized the ESI handbook's 150 training cases and their correct answers, then studies reporting high LLM triage accuracy on those same cases are measuring recall of training data, not clinical competence. The 70.6% accuracy reported by Pash et al. (2024) using 100 ESI handbook cases, substantially outperforming their emergency physician panel at 57.3%, may partially reflect this confound.

Fourth, the scoring framework is wrong. Existing benchmarks score all errors equivalently. A model that assigns ESI Level 3 to a true Level 2 patient (a one-level undertriage that delays bedside evaluation of a potentially unstable patient) receives the same penalty as a model that assigns Level 2 to a true Level 3 patient (a one-level overtriage that accelerates evaluation of a stable patient). In clinical practice, these errors are not equivalent. The undertriage may result in a patient with an evolving stroke sitting in a waiting room while the treatment window closes. The overtriage results in a faster-than-necessary evaluation of a patient who was going to be fine either way. The only published LLM triage study to implement asymmetric scoring — Gaber et al. (2025) in *npj Digital Medicine* — found that this directional weighting materially changed the comparative performance rankings of the models tested. The NOHARM benchmark's finding that errors of *omission* accounted for more than 80% of severe-harm potential across systems (Wu et al., preprint) is the consultation-side confirmation of the same principle: the dangerous error class is what a system fails to say, and symmetric scoring cannot see it.

These four structural incompatibilities — format, construct, contamination, and scoring — are not correctable through incremental modification of existing benchmarks. They require a purpose-built evaluation instrument designed from its foundations for the specific task of triage safety assessment.

The Closest Prior Work — and What It Leaves Unbuilt

In February 2026, the first product-level evidence arrived. Ramaswamy and colleagues at Mount Sinai published in *Nature Medicine* a structured stress test of ChatGPT Health using 60 clinician-authored vignettes across 21 clinical domains under 16 factorial conditions (960 responses). The results independently confirm this protocol's central premise. Performance followed an inverted-U pattern, failing most often at the clinical extremes: 48% on emergency conditions and 35% on nonurgent presentations. Among gold-standard emergencies the system undertriaged 51.6% of cases — directing presentations such as impending respiratory failure to 24–48-hour evaluation rather than the emergency department — while triaging classical, pattern-complete emergencies such as stroke and anaphylaxis correctly. When family members minimized symptoms, recommendations shifted toward less urgent care in edge cases (OR 11.7, 95% CI 3.7–36.6). Crisis-intervention safeguards activated unpredictably across suicidal-ideation presentations, firing more often when no method was described than when one was. This is the disposition-failure phenotype — delayed escalation under benign framing — registered under controlled conditions in the deployed product.

What Ramaswamy et al. establish is the danger; what they leave unbuilt is the standard. Their evaluation tested a single product at a single timepoint, with one response per vignette-condition cell, so run-to-run nondeterminism was not measured. Triage was elicited as a forced single-letter choice on a four-level scale whose ceiling is “go to the emergency department,” without distinguishing EMS activation from self-transport — the boundary at which safety-critical instructions (“call 911; do not drive”) operate, and the boundary at issue in *Winters*. The gold standard was assigned by three physician-investigators drawn from urology and internal medicine, with no independent physician answering arm responding under the model's informational constraints. Presentation register was not an experimental factor. Errors were tallied without severity weighting, pre-specified killer items, or time-to-escalation. Contamination was mitigated by novel authorship but not directly audited. And — as the authors themselves note — no human-AI interaction was tested.

Within weeks, the finding itself became contested. A partial replication (Fraile Navarro, Magrabi and Coiera; preprint) re-ran seventeen of the scenarios on five frontier models under two conditions — the original forced-choice format and naturalistic free-text patient messages — and

found the forced format to be a strong failure mechanism in its own right: on matched scenarios, models that scored as low as 0–24% under forced choice recommended emergency care in their own words 100% of the time, and naturalistic interaction improved accuracy by 6.4 percentage points overall. The replication is itself limited — it substituted newer API models for the actual product, and clean single-turn vignettes cannot capture the under-reporting and misapplication that degrade real consumer interactions, as Bean et al.’s randomized evaluation demonstrates. Two further preprints sharpen the dispute. When the identical benchmark was applied to OpenEvidence, a physician-facing product, emergency undertriage fell four-fold to 12.5% with no anchoring effect — under the same forced-choice format, so elicitation alone does not determine the failure profile; product and configuration matter (Jia et al., preprint). And when ChatGPT Health itself was granted multi-turn conversation, disposition alignment did not materially improve, with roughly 70% of discordant recommendations still under-triaged (Tyagi et al., preprint — from the original Mount Sinai group). The field is therefore left with a public methodological dispute: product-level failure or evaluation-format artifact, with the true deployed behavior unresolved. The present protocol is designed to adjudicate exactly this question. It elicits free-response recommendations and scores explicit escalation deterministically from the model’s own words; it manipulates presentation register as a crossed factor; it evaluates multiple version-pinned models with repeated runs; it weights errors by severity against an independent emergency-physician benchmark; and it tests real human-AI interaction. Each of these choices resolves one axis of the dispute the existing literature cannot.

The same year produced the closest clinician-facing analog. NOHARM (Wu, Nateghi Haredasht et al., preprint) built an 1,100-task consultation-safety benchmark from real electronic-consult cases, with 29 physicians pre-annotating 4,249 management options for benefit versus harm on a RAND/UCLA appropriateness scale with WHO harm-severity definitions, and paired it with a 101-physician randomized teaming study. Its construct is consultation comprehensiveness for clinicians — not consumer disposition safety — and its authors’ stated limitations (single-turn text, no patient-facing register, no interaction testing) mark the boundary where the present protocol begins. The 2026 landscape thus holds three benchmark efforts: NOHARM on the clinician-facing side; Ramaswamy et al.’s consumer product stress test, now contested on measurement grounds; and the instrument proposed here — the consumer-disposition safety standard with an independent physician benchmark — which remains unbuilt. Ramaswamy et al. close by arguing that consumer triage AI “should not be deployed on trust alone”; their critics answer that valid evaluation “requires testing under conditions that reflect actual use.” Both statements are arguments for this instrument.

Study Design: A Physician-Authored, Contamination-Resistant Triage Benchmark

This study constructs that instrument. Fifty novel emergency triage scenarios will be authored de novo by emergency physicians at Cedars-Sinai Medical Center, derived from real emergency department practice patterns and structured using the Medical Council of Canada's Key Features Question methodology — the most validated written assessment of clinical decision-making in medical education, with over thirty years of psychometric evidence supporting its validity.

Scenario Architecture. Each scenario is a composite clinical case limited to information available at the triage stage: chief complaint, age, sex, relevant past medical history, vital signs, and clinical appearance. No laboratory results, imaging findings, or diagnostic workup data are included, because these do not exist at the moment of triage. Each scenario exists in two parallel formulations. The lay register presents the clinical information as a patient would describe it to a chatbot — first-person, everyday language, incomplete, sometimes misleading, using phrases like “my stomach feels off” rather than “I have epigastric discomfort.” The expert register presents the same clinical information as a structured triage vignette with standardized vital sign reporting and clinical terminology. Both formulations contain sufficient information for correct triage; they differ in register, not in clinical content. This dual-formulation design tests a critical deployment question: does the model's triage performance degrade when clinical information is embedded in the imprecise, ambiguous language that real patients actually use? It is also, after the events of 2026, the design feature that arbitrates between “the product fails” and “the format failed the product.”

Key Features Methodology. Each scenario contains two to three pre-defined key features — the specific clinical decision points where triage errors cluster. These are not “important clinical facts” but rather the steps where clinicians are most likely to go wrong, following Bordage and Page's foundational distinction. For a case of atypical acute coronary syndrome in an elderly diabetic patient, the key features are not “chest pain suggests MI” (which every clinician and model gets right) but rather “absence of chest pain does not rule out MI in elderly diabetic patients” (where errors concentrate), “appropriate urgency requires ED evaluation now, not home observation” (where the disposition decision is made), and “explicit instruction not to drive and not to wait” (where safety-critical communication occurs). Each key feature maps to a specific sub-question in the evaluation template, enabling granular assessment of where in the clinical reasoning chain a model succeeds or fails. Ramaswamy et al.'s central pattern — classical presentations triaged correctly while atypical, pattern-incomplete presentations were missed — is the key-features thesis observed in deployed-product data.

Killer Items. Each scenario specifies two to three responses that constitute automatic critical failures — the dangerous errors that could directly lead to patient death or irreversible harm. These are pre-defined before any model is evaluated, eliminating post hoc interpretation. For a case of thunderclap headache suspicious for subarachnoid hemorrhage, the killer items are: attribution to migraine without recommending immediate evaluation, recommendation to take NSAIDs and rest, and failure to specify emergency transport. A model that correctly identifies the headache as concerning but tells the patient to “drive to the nearest emergency room when you feel up to it” has committed a safety-critical failure that would not be captured by any existing triage benchmark’s scoring rubric.

Severity Weighting. Each scenario carries a severity weight of 1, 2, or 3, corresponding to the time horizon to serious harm if the presentation is undertriaged. Weight 3 denotes minutes-to-death presentations (subarachnoid hemorrhage, massive pulmonary embolism, ruptured ectopic pregnancy). Weight 2 denotes hours-to-severe-harm presentations (sepsis, NSTEMI, diabetic ketoacidosis). Weight 1 denotes days-to-weeks presentations where delayed care carries meaningful but less immediate risk. The Failure Severity Index aggregates errors weighted by these severity classifications, ensuring that a missed PE carries substantially more weight in the final analysis than a missed mild presentation. Harm-severity determinations by the scoring panel are anchored to the RAND/UCLA appropriateness methodology and WHO International Classification for Patient Safety severity definitions, as operationalized in the NOHARM benchmark; a response is assigned the highest harm tier only on unanimous rater agreement, and is otherwise downgraded one tier — a deliberate conservatism against noisy severe-harm attribution.

Configuration and Cue Factors. Two crossed factors probe deployment-relevant failure modes. A configuration factor compares each model’s default consumer behavior against a minimally scaffolded configuration requiring an explicit “Urgent Medical Attention” assessment after any differential (Amendment 1). A matched-pair bias-cue factor manipulates the locus of benign framing — no cue, note-embedded cue (e.g., a triage impression of “history of anxiety, likely panic attack”), or user-asserted cue (e.g., a family member minimizing symptoms) — across anchoring, social-pressure, base-rate, availability, and affect cue classes, with one over-ordering cue as a directional control (Amendment 2). Ramaswamy et al.’s family-minimization result (OR 11.7, significant in edge cases only) establishes the conversational locus empirically; the note-embedded locus remains unexamined in any published study. Cue effects are analyzed as within-scenario matched pairs, which are robust to overall evaluation-format effects in a way absolute rates are not.

Contamination Resistance. Because no existing published benchmark can be trusted as uncontaminated for frontier models, the entire 50-scenario

instrument is authored de novo from clinical experience. The scenarios are composite cases that have never been published, posted, or digitized anywhere. They use compositional complexity — combining known conditions in novel configurations that resist pattern-matching — and the dual-register formulation itself provides additional protection, since even if the underlying clinical entity is recognizable, the specific lay-language presentation will not match any training data verbatim. Embedded contamination detection items — a subset of minimally modified known cases alongside the novel composites — will provide direct evidence of whether contamination effects are present, rather than assuming they are absent as every prior study has done. For eventual release of the instrument, the study will follow the public/private-split stewardship model demonstrated by NOHARM, including a deliberate-overfitting probe that shows contamination of released cases is detectable.

The Three-Arm Evaluation Architecture

The study employs three evaluation arms that generate directly comparable data. The architecture follows from a principle now stated explicitly in the medical AI safety literature: a system's safety class is "not a property of the AI model alone... It depends on how the model is deployed, including the clinical use, clinical workflow, patient population, availability of follow-up, and the clinician's ability to override the recommendation" (Sheng et al., *NEJM AI*, 2026 — a proposed framework). A model-alone benchmark number is therefore an incomplete safety claim by construction — not because the measurement is imprecise, but because safety is a property of the deployed configuration rather than of the weights. Arms A and C measure two configurations of the same models under the same scenarios, and the distance between them is itself a primary finding rather than a robustness check.

Arm A: Model Alone. Each frontier large language model — including but not limited to the current versions of ChatGPT, Claude, Gemini, and their successors — responds to all 50 scenarios in both lay and expert registers using a standardized prompt template. The template elicits four components: an ESI-equivalent triage level assignment (1 through 5), identification of key clinical features driving the triage decision, specification of immediate recommended actions, and a brief disposition rationale explaining why the patient does or does not need immediate physician assessment. Responses are free text throughout; no answer options are ever presented. Each scenario is run at least three times per model per condition, with model identifier, version, access date, temperature, and all sampling parameters pinned and reported; the majority call is primary and run-to-run agreement is reported as an instability metric, so that nondeterminism is measured rather than averaged away. Models are accessed via API without web or tool access, preventing live benchmark identification or answer retrieval during evaluation.

Arm B: Physician Benchmark. An independent physician answering panel — emergency physicians who did not participate in scenario authoring — responds to the same scenarios under identical informational constraints. They see the same vignettes, answer the same four-component template, and have no access to the gold standard or to each other’s responses. This arm establishes a true human comparator baseline. The strict separation between authoring and answering panels eliminates authorship familiarity bias — the tendency for case authors to perform artificially well on their own cases — which would inflate the physician benchmark and unfairly disadvantage the AI comparator. No published LLM triage study has implemented this separation. In the most recent ChatGPT Health evaluation, the physicians who set the gold standard were study investigators, and no independent physician panel answered the scenarios under the model’s informational constraints; the human role was adjudication only. The present design separates authoring, gold-standard determination, answering, and scoring into four non-overlapping panels.

Arm C: Human-LLM Interaction. Lay participants interact with an LLM before making their own triage decisions, simulating real-world deployment conditions. This arm captures interaction failure modes that neither model-alone nor physician-alone testing can reveal: information omission (the user fails to communicate key clinical information), model misinterpretation (the model produces unsafe reasoning despite adequate input), suggestion adoption failure (the user does not follow correct model recommendations), and interaction variability (small changes in phrasing produce meaningfully different model outputs). The NOHARM randomized study has now demonstrated suggestion adoption failure in physicians — attending clinicians frequently omitted appropriate actions their own AI assistant had surfaced; this arm asks whether lay users do the same under the conditions that matter most. The consultation order is a deliberate design choice and is stated as such: participants consult the model *before* forming their own disposition, because that is the order in which real patients use these systems. Order is not neutral — in human-AI drafting studies, whichever party responds second tends to defer to the first, so anchoring runs in both directions depending on sequence. Two consequences are pre-specified. First, agreement between a participant and a correct model recommendation is not by itself evidence of independent judgment, and the interaction metrics are interpreted accordingly. Second, an exploratory counterbalanced condition — in which a subset of participants records an initial disposition before consulting — is included to estimate the magnitude of the order effect directly rather than leaving it as an uncontrolled feature of the design. These metrics address a dimension of AI safety that is entirely absent from benchmark-based evaluation: the gap between what a model can do in isolation and what actually happens when a frightened patient communicates with it at two in the morning.

Scoring and Adjudication

The scoring framework implements three design principles that distinguish it from all prior work: asymmetry, determinism-first automation, and structured adjudication with humans sovereign over harm.

Asymmetric Scoring. Following Gaber et al.’s triage range model, the scoring rubric penalizes undertriage more heavily than overtriage. Exact match with the gold standard ESI level earns full credit. One-level overtriage — assigning a more conservative level than the gold standard — earns substantial partial credit, reflecting the clinical reality that a patient evaluated faster than strictly necessary suffers minimal harm. One-level undertriage — assigning a less conservative level — earns minimal credit, reflecting the clinical reality that a patient whose evaluation is inappropriately delayed faces genuine risk. Discrepancies of two or more levels in either direction earn zero credit, as these represent categorically unsafe performance regardless of direction. Killer items trigger automatic zeros for the entire key feature they relate to, regardless of other correct elements in the response.

Determinism-First Hybrid Evaluation. The scoring pipeline routes each criterion to the least interpretive method capable of scoring it. Approximately 40% of the total scoring weight — the explicit-escalation determination, ESI level match, key feature checklist items, killer item detection, and format compliance — is scored mechanically. In particular, explicit escalation (the primary endpoint’s core element) is detected by a pre-registered, deterministic phrase-inventory rule set applied to the model’s own free-text words, validated against a physician-labeled sample before unblinding; no language model judges the primary endpoint. Approximately 30% — differential diagnosis completeness and reasoning coherence against reference rationales — is evaluated by an ensemble of at least three independent frontier LLM judges from different vendors, version-pinned, with the majority verdict used. The ensemble is subordinate and provisional: it is accepted for each component only if its agreement with the blinded physician panel on a pre-registered validation sample is at least as high as inter-physician agreement on the same items — the acceptance logic established by NOHARM, whose extract-and-match autograder reached $\kappa = 0.804$ against physicians versus inter-physician $\kappa = 0.784$ — and only if chance-corrected agreement meets the pre-registered threshold ($\text{Gwet AC1} \geq 0.75$). Per-judge self-preference diagnostics (whether a vendor’s judge scores that vendor’s outputs more favorably) are reported. Any component failing these gates reverts entirely to physician scoring. The remaining 30% — harm potential assessment, reasoning quality in ambiguous cases, communication appropriateness — is reviewed by the physician scoring panel and is never delegated to models. This tiered approach is both a practical necessity (with 50 scenarios, 6 or more models, 3 to 5 runs each, and multiple crossed conditions, the study generates

thousands of individual responses) and a methodological answer to the model-judging-model objection.

Three-Tier Adjudication. For the human-evaluated components, the study implements a structured adjudication protocol drawn from the ClinicalKey AI evaluation (Livingston et al., 2025, *JAMIA Open*), the most operationally detailed published workflow for clinical AI scoring. Each response is independently rated by two blinded physician raters. If both agree across all scoring dimensions, the scores are accepted as final. When any disagreement exists, a third rater independently evaluates the same response, and the mode across three ratings becomes the final score. For persistent three-way disagreements where no mode exists, raters convene for a structured modified Delphi discussion — each presents their reasoning, and discussion continues until consensus or supermajority is reached. This escalation ladder resolved over 60% of cases at the first tier in the ClinicalKey study, minimizing the total adjudication burden while ensuring no contested score goes unresolved.

Statistical Analysis

The primary analysis compares critical miss rates across models and between AI and physician panels using mixed-effects logistic regression. The model specification includes fixed effects for AI system, presentation register (lay versus expert), configuration, and evaluation mode (model-alone versus human-LLM interaction), with scenario treated as a random effect to account for case-level clustering. A model-by-register interaction term tests whether specific AI systems show differential vulnerability to lay-language presentations — a finding with direct implications for deployment safety, since real-world use is overwhelmingly in the lay register.

Two mechanical reference policies are reported alongside every model and the physician panel: a “never escalate” floor and an “always escalate” ceiling. The floor quantifies the harm of reflexive reassurance; the ceiling quantifies the cost of indiscriminate referral (the universal-referral rate is reported per configuration so that any scaffold benefit reads as an operating-point improvement rather than blanket escalation). A deployable system must beat both simultaneously; this is the disposition-domain analog of NOHARM’s “do nothing” comparator, which carried severe-harm potential in 37% of consultation cases.

Multiple comparisons across six or more AI models are corrected using the Benjamini-Hochberg false discovery rate procedure. Inter-rater reliability is reported with Gwet’s AC1 as the primary agreement statistic (weighted kappa secondary, as kappa is unstable at the skewed prevalence expected for safety-critical tags), and Krippendorff’s alpha for the full rater panel. Bias-cue contrasts are analyzed as within-scenario matched pairs (McNemar’s test; conditional logistic regression for interactions) under

hierarchical gatekeeping; no power figures are stated for cue analyses until pilot discordance data exist. Secondary analyses include the Failure Severity Index (aggregating severity-weighted errors), explicit-escalation rate, time-to-escalation (the token or sentence position of the first explicit escalation recommendation), anchoring failure rate (the proportion of cases where the model attributes a dangerous presentation to a benign cause), run-to-run instability, crisis-safeguard activation consistency across matched suicidality presentations (exploratory, extending Ramaswamy et al.'s inverted-activation finding), and level-specific accuracy via confusion matrices that separately report undertriage and overtriage rates at each ESI level boundary.

The study's power derives from the number of scenarios (50 cases at 2 to 3 key features each, yielding approximately 120 to 150 individual scored items), the number of repeated runs per scenario per model (3 to 5, providing within-model variability estimates), and the crossed design (each model sees every scenario in both registers). With 50 cases, the MCC's empirical research predicts Cronbach's alpha of approximately 0.70 to 0.80 — the threshold the Medical Council of Canada found sufficient for a high-stakes national licensing examination with 32 cases.

What This Study Does Not Establish

Three boundaries are stated here rather than left for reviewers to find.

This is an encounter-time instrument measuring a hazard with a delayed tail. The deployment safety classification recently proposed for medical AI separates errors a clinician can catch inside the ordinary workflow (S2) from errors that “may not be visible during the encounter” and whose harm “depends on missed follow-up, delayed diagnosis, or long-term management” (S3) — listing among its S3 examples AI tools that affect referral urgency and diagnostic timing (Sheng et al., *NEJM AI*, 2026). Triage advice is such a tool. The Critical Miss Rate is scored at the encounter: deterministically, from the model's own words, at the moment of response. It is therefore an S2-grade instrument applied to a hazard with an S3 tail. A model posting a low critical miss rate here has not been shown to leave patients unharmed months later; it has been shown that its encounter-time output was not refuted against a physician-defined gold standard. The framework's own epistemic formulation is the one to adopt: absence of observed harm means only that safety has not been refuted. Establishing S3-level safety would require linking AI-influenced dispositions to downstream outcomes — the multiwindow surveillance design (30-, 90-, and 365-day checkpoints, with registry or claims linkage beyond twelve months) that the same authors propose. That is a successor study. This protocol is deliberately the predeployment layer beneath it, and claims nothing above that layer.

Vignettes are not patients. Composite scenarios authored by emergency physicians hold clinical content constant across models, registers, and runs, which is precisely what makes model-to-model and register-to-register comparison valid. They do not reproduce the information loss, emotional pressure, interruption, and partial disclosure of a frightened person describing symptoms in real time. Arm C narrows this gap by placing lay participants in the interaction loop; it does not close it, and no vignette-based design can.

A benchmark result is a measurement, not a clearance. Nothing in this design licenses a deployment decision on its own. The study produces a comparative safety measurement under specified conditions against pinned model versions, and that measurement begins decaying the moment a vendor ships a new generation — which is why the deliverable is a reusable instrument rather than a verdict on any particular system.

Expected Findings and Their Consequences

Several patterns of results are anticipated, each carrying distinct implications.

Models will likely perform well in the expert register and substantially worse in the lay register. Structured clinical presentations with standardized vital sign reporting, medical terminology, and organized history elements align closely with the clinical text on which frontier models were trained. Lay-language presentations — with their ambiguity, incompleteness, emotional coloring, and absence of clinical framing — present a fundamentally harder input processing challenge. If this performance gap is confirmed, it directly undermines the safety of current consumer-facing deployment, since patients do not present symptoms in clinical language. The magnitude of this gap — whether it is 5 percentage points or 25 — will determine the urgency of the policy response. Because both registers are scored from free-response answers, any gap found here cannot be attributed to elicitation format — the objection that now shadows the constrained-format literature.

Undertriage errors will cluster at specific clinical boundaries. Based on the published triage literature, the most error-prone boundary is ESI Level 2 versus Level 3 — the distinction between high-risk patients requiring immediate bedside evaluation and stable patients who can wait for care. Kaiser Permanente data suggests approximately 25% of all triage assignments at Level 2 represent overtriage, and the Level 2/3 boundary accounts for the largest share of inter-rater disagreement among human clinicians. Models may show a similar pattern, or they may show a different error distribution — either finding is informative. If models undertriage at the Level 1 boundary (failing to recognize immediate life threats), the safety

implications are more severe than if they overtriage at the Level 3/4 boundary (overestimating resource needs for stable patients).

Anchoring failures will be disproportionately dangerous. Several scenarios in the benchmark are specifically designed to include benign-sounding anchors — a migraine history in a patient with thunderclap headache, a recent normal endoscopy in a patient with aortic dissection, an anxiety diagnosis in a patient with massive pulmonary embolism. If models latch onto these anchors and attribute dangerous presentations to benign causes, this represents the most clinically consequential failure mode: confident false reassurance. This prediction now has direct empirical support: in the deployed product, third-party minimization of symptoms shifted triage toward less urgent care with an odds ratio of 11.7 — and only in boundary cases, precisely where this benchmark’s scenarios are deliberately targeted. The note-embedded cue locus — benign framing arriving in the clinical record rather than the conversation — remains unexamined in any published study and is tested here. A patient told “this sounds like your usual migraine — take your rescue medication and rest in a dark room” when they are actually bleeding into their subarachnoid space will not seek further care. The anchoring failure rate may prove to be the single most actionable metric in the study.

Human-LLM interaction will reveal failure modes invisible to benchmark testing. The interaction arm is expected to show that real-world triage safety depends not just on model competence but on communication dynamics: whether patients volunteer critical information unprompted, whether models ask the right follow-up questions, whether patients follow correct recommendations, and whether semantically similar inputs produce consistent outputs. If a model correctly identifies a high-risk presentation in 4 of 5 runs but misses it in the fifth due to minor phrasing variation, that stochastic failure has patient safety implications that no single-run benchmark evaluation can capture.

Consequences for Clinical Practice and Policy

The first product-level measurements now exist — and they simultaneously confirm the danger and expose the measurement problem. What still does not exist is the standard-setting instrument: a multi-model, physician-benchmarked, severity-weighted, contamination-audited, free-response, interaction-inclusive evaluation that can be reapplied as models change. This study supplies it.

If models demonstrate acceptable critical miss rates across both registers and all severity levels, that finding provides the first evidence-based basis for cautious optimism about consumer health AI in acute settings. If they demonstrate unacceptable miss rates — particularly in the lay register, at high severity weights, or in anchoring-vulnerable scenarios — that finding

creates an evidence-based mandate for clinical warning, patient counseling, regulatory attention, and engineering improvement. Either finding is valuable, and either finding will stand on methods that both sides of the current measurement dispute have implicitly endorsed: free-response elicitation under conditions that reflect actual use, scored against a physician gold standard with the severity of harm made explicit.

These findings will land inside an active regulatory architecture rather than a vacuum. The European Union's Artificial Intelligence Act imposes post-market monitoring obligations under Article 72; the U.S. Food and Drug Administration's final guidance on predetermined change control plans (August 2025) governs how AI-enabled device software may be updated after authorization; the International Medical Device Regulators Forum has issued software-specific risk characterization (N81) and good machine learning practice (N88) guidance; and the National Institute of Standards and Technology's AI Risk Management Framework and World Health Organization guidance on large multi-modal models supply complementary governance vocabulary. What none of these instruments supplies is the measurement. Consumer triage AI largely escapes device regulation by positioning itself as wellness information rather than as software as a medical device, and the frameworks that would otherwise govern it are calibrated to evidence that does not yet exist for this use case. A physician-benchmarked critical miss rate — severity-weighted, contamination-audited, and measured in the register patients actually use — is the evidentiary object those frameworks require and currently lack.

What is not acceptable is the current state: widespread deployment of patient-facing AI in a triage-critical role, a headline safety estimate contested on methodological grounds within weeks of publication, and no instrument capable of resolving the question. This study replaces that vacuum with measurement. It establishes a reusable benchmark instrument — a 50-scenario, physician-authored, dual-register evaluation framework with pre-defined key features, killer items, and severity weights — that can be applied to future models as they are released, creating a longitudinal safety record for patient-facing AI in emergency medicine. It positions the evaluation standard where it belongs: in the hands of the emergency physicians who define the boundary between safe discharge and life-threatening delay, rather than in the hands of the AI developers whose systems are being evaluated.

The clinical question this study addresses is simple enough for any patient to understand and consequential enough for any institution to prioritize: *Can I trust this AI when it tells me I don't need to go to the emergency room?* The answer requires measurement. This protocol provides it.

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Verified against PubMed/Scite as of August 5, 2026; reference 18 added and verified against the publisher record August 21, 2026. Preprints are identified as such; per protocol citation policy, preprint findings are presented with that caveat and re-verified upon journal publication.

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